

Exercise 2: E-commerce Platform Search Function

Product.java

```
public class Product {  
  
    int productId;  
  
    String productName;  
  
    String category;  
  
    public Product(int id, String name, String category) {  
  
        this.productId = id;  
  
        this.productName = name;  
  
        this.category = category;  
    }  
  
    public String toString() {  
  
        return "[" + productId + "] " + productName + " (" + category + ")";  
    }  
}
```

SearchDemo.java

```
import java.util.Arrays;  
import java.util.Comparator;  
  
public class SearchDemo {  
  
    public static void main(String[] args) {  
  
        Product[] products = {  
  
            new Product(101, "Laptop", "Electronics"),  
  
            new Product(205, "Shoes", "Fashion"),  
  
            new Product(150, "Phone", "Electronics"),  
  
            new Product(303, "Watch", "Accessories"),  
  
            new Product(175, "Bag", "Fashion")  
        }  
    }  
}
```

```

};

int targetId = 150;

System.out.println("Linear Search:");
Product result1 = linearSearch(products, targetId);
System.out.println(result1 != null ? result1 : "Product not found");

Arrays.sort(products, Comparator.comparingInt(p -> p.productId));
System.out.println("\nBinary Search (after sorting):");
Product result2 = binarySearch(products, targetId);
System.out.println(result2 != null ? result2 : "Product not found");
}

```

```

static Product linearSearch(Product[] items, int targetId) {
    for (Product item : items) {
        if (item.productId == targetId) return item;
    }
    return null;
}

```

```

static Product binarySearch(Product[] items, int targetId) {
    int left = 0, right = items.length - 1;
    while (left <= right) {
        int mid = (left + right) / 2;
        if (items[mid].productId == targetId) return items[mid];
        if (items[mid].productId < targetId) left = mid + 1;
        else right = mid - 1;
    }
    return null;
}

```

```
}
```

Output :

```
PS C:\Abishek Khanna\Projects\Java_FSE_Deepskilling_CTS\Week1_Algorithm_Data_Structure> & 'C:\Program F3
' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\abish\AppData\Roaming\Code\User\workspaceStor
\Week1_Algorithm_Data_Structure_9e30e213\bin' 'SearchDemo'
Linear Search:
[150] Phone (Electronics)

Binary Search (after sorting):
[150] Phone (Electronics)
PS C:\Abishek Khanna\Projects\Java_FSE_Deepskilling_CTS\Week1_Algorithm_Data_Structure>
```

Exercise 7: Financial Forecasting

FinancialForecast.java

```
import java.util.Scanner;
```

```
public class FinancialForecast {
```

```
    public static double forecastValue(double presentValue, double growthRate, int years) {
```

```
        if (years == 0) {
```

```
            return presentValue;
```

```
        }
```

```
        return forecastValue(presentValue, growthRate, years - 1) * (1 + growthRate);
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter present value (in rupees): ");
```

```
        double present = sc.nextDouble();
```

```

System.out.print("Enter annual growth rate (in percent): ");

double ratePercent = sc.nextDouble();

double growthRate = ratePercent / 100;


System.out.print("Enter number of years: ");

int years = sc.nextInt();


double future = forecastValue(present, growthRate, years);

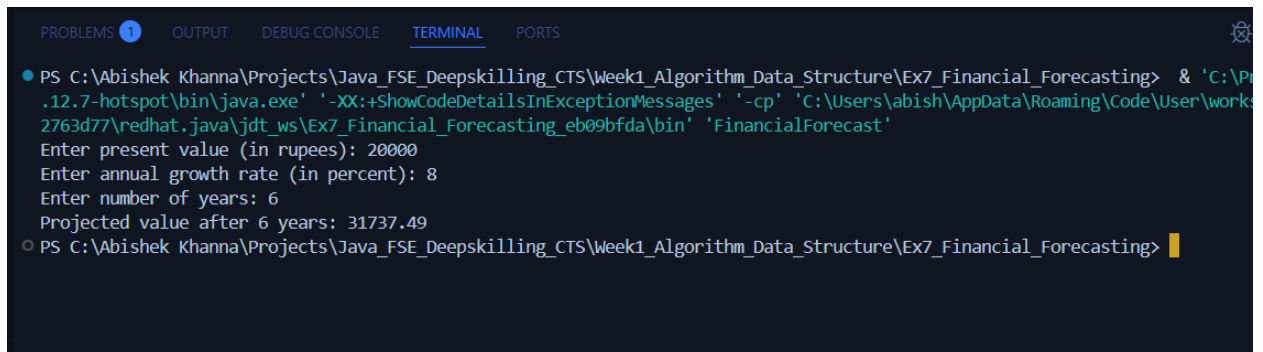

System.out.printf("Projected value after %d years: %.2f\n", years, future);

}

}

```

Output :



The screenshot shows a terminal window with the following content:

```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Abishek Khanna\Projects\Java_FSE_Deepskilling_CTS\Week1_Algorithm_Data_Structure\Ex7_Financial_Forecasting> & 'C:\Program Files\Java\jdk-12.0.2\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\abish\AppData\Roaming\Code\User\workspace\2763d77\redhat.java\jdt_ws\Ex7_Financial_Forecasting_eb09bfda\bin' 'FinancialForecast'
Enter present value (in rupees): 20000
Enter annual growth rate (in percent): 8
Enter number of years: 6
Projected value after 6 years: 31737.49
○ PS C:\Abishek Khanna\Projects\Java_FSE_Deepskilling_CTS\Week1_Algorithm_Data_Structure\Ex7_Financial_Forecasting>

```